# *Automated signature inspection and forgery detection utilizing VGG-16: a deep convolutional neural network*

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*Abstract*— With the growing usage of handwritten signatures in legal documents and financial transactions, it is crucial to select an efficient technique for validating these signatures and eliminating forgeries that might cause considerable losses for clients. Offline signature authentication is yet challenging despite extensive study, notably when trying to separate expert forgeries from signature pools. This research aims to develop an automated system using VGG-16 a deep convolution neural network (DNN) to successfully tackle handwritten signature fraud. The automated system is developed separately by utilizing a pre-trained and a trainable VGG-16, whose performance is accessed over a widely used signature dataset comprising an equal number of authentic and fake signatures. In comparison to the pre-trained VGG-16 model, the trainable VGG-16 based model can yield a testing accuracy of 0.962, which is about 17.75% higher. The simulation outcomes decisively illustrate that the recommended methodology is suitable for automatically spotting fraud handwritten signature images.

***Keywords***— CNN, VGG-16, Classification, forgery identification

## I. INTRODUCTION

The use of signatures is a common way for individuals to formally bind a document or express their preference for specific information, and is frequently used in financial and legal activities such as bank letters, property documents, checks, and corporate documents [1]. However, signature forgery has become a major cybercrime, with individuals attempting to duplicate signatures for financial gain. In recent years, the rapid growth of technology has made signature forgery more sophisticated, making it difficult to detect with traditional methods. As a result, advanced techniques such as deep learning algorithms are necessary to accurately identify forged signatures.

A variety of forgery detection systems built on deep learning has been presented in the literature. In [1], a straightforward CNN including three convolution and maxpooling layers setup alternatively with one flattened and dense layer is suggested for spotting forgeries of signatures. The author created a dataset with 6000 signatures with three persons. In order to be fed into the CNN, originally collected RGB format signatures are first transformed to grey scale and then to bitmap format. For model evaluation, accuracy and loss percentage metrics ar e utilized. Two DNNs, one developed using transfer learning and another customized one are compared with traditional SVM and KNN techniques for signature recognition in [2]. Clearly, the DNN models beat the conventional ones when results from two DNNs were compared over a dataset of 600 signatures generated from 30 persons. In [3], the author suggests a CNN model for forgery detection that combines the Harris-Surf and Crest-Trough algorithms. The suggested CNN model architecture is composed of 5 layers, and the model is trained using the back-propagation technique. The CEDAR dataset, which has 1320 photos for each authentic and fake signature, is used to test the model. Siamese neural network a variety of DNN is suggested by the authors of [4] as a possible solution to signature verification problem. Transforming to grey scale, image scaling, morphological erosion and augmentation are used as part of pre-processing stage. The model performance is then compared with LeNet and SigNet on an artificially created signature dataset. In [5], another DNN is proposed for signature verification that uses the ResNet-50 with some more layers and encoders. A Kaggle dataset having 2660 English signatures is used in the model's testing. The use of Mutual Signature DenseNet (MSDN) in a region-based DNN is investigated in [6]. It is made up of two identically structured CNN branches with same learnable parameters. The MSDN feature extractor is used in their suggested technique to analyze numerous local regions, and the similarity between two signatures is then used to train a Siamese Neural Network (SNN). The model is then evaluated by comparing it to several CNN architectures, including VGG-16, AlexNet, and ResNet-18. A tailored four layered CNN optimized using ADAM optimizer has shown a validation accuracy of 88.39% over a synthetic dataset with 770 signatures prepared from 55 persons each one contributing 7 genuine and 7 forged one in [7]. The authors of [8] have experimented with a CNN model considering various split ratios of datasets to identify fake signatures. Accuracy and

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loss are considered as the evaluation criteria in their study. After creating a handwritten signature dataset using attendance records, the authors of [9] have utilized a feed forward neural network to differentiate between signatures and non-signatures. A deep CNN classifier that had been separately trained using the ADAM and SGD optimizers and utilizing various hyper-parameter tweaking techniques was then used to confirm the validity of the signature. A tailored CNN designed in [10] to identify fake signatures outperformed a VGG-16, a SNN and a SNN with signet architecture in terms of accuracy when tested on the CEDAR dataset. Another tailored CNN employing 6 convolutions with 5 maxpooling, 1 flatten and 2 dense layers is presented in [11] to identify signature forgery. The model is evaluated over a sample of 300 photos with an equal percentage of authentic and fraudulent signatures from 30 persons. A custom CNN validated with 130 signatures collected from 130 individuals has shown accuracy more than 95% accuracy during validation and aw well as when tested with 20 images in [12].

The goal of this project is to develop a simple VGG-16 based automated signature verification system (ASVS) that can identify whether a given signature is genuine or fraudulent. With given a signature image as input, the model will use the various convolution layers of VGG-16 to extract essential features and then utilizing these features will classify the signature at the output layer. Taking the publicly available CEDAR dataset as a reference, the performance of ASVS modeled with a trainable VGG-16 and with a pre-trained VGG-16 is accessed with respect to four assessment metrics, such as accuracy, precision, recall, and F1-score. The simulation outcomes convincingly indicate the viability of the proposed technology in recognizing forgery in the handwritten signature visuals.

The working mechanism of the ASVS is explained in the next part. Section 3 analyzes the outcomes followed by conclusion in next paragraph.

## II. ASVS USING VGG-16

Fig. 1 depicts the overall working principle of the VGG-16 based ASVS. To assess the effectiveness of the proposed, a standard database of signatures is required. For this purpose, publicly available handwritten signatures related to 55 persons are collected from https://cedar.buffalo.edu/NIJ/data. To ensure that that the performance of the system should not be biased by the size and slant offered in the signature, the collected signature images need pre-processing. So in the first step of data pre-processing, to attain homogeneity across the images, which ranged from 200px to 800px in size, we adopted OpenCV's resize function setting the interpolation parameter as INTER_LINEAR and rescaled the images to a standard resolution of 250px to 250px prescribed for the VGG-16 model. After that, noise removal from images is accomplished utilizing a Gaussian blur filter. The fastNlMeansDenoising function of the OpenCV is used to apply the filter. Following the pre-processing, a predefined splitting ration was used to divide the dataset into training and testing sets.

Then a VGG-16 based DNN developed using transfer learning is utilized for classification of the signature images as genuine or forged. Numerous studies have been inspired by VGG-16 to utilize it to address a range of object detection and classification problems [2][7][12]. It has already shown 92.7% accuracy to classify 1000 images of 1000 different categories. The 13 convolution layers, 5 maxpooling layers and 3 dense layers form the 21 layers of VGG-16, but 16 of them are weight layers with learnable parameters. The network has total 14,764,866 parameters out of which 50,178 are trainable parameters and remaining 14,714,688 are non-trainable ones. Convolution produces a fixed number of feature maps from the input image, which are then fed into a max pooling layer to produce pooled feature maps. The subsequent convolution layer receives these pooled feature maps, and so on until we attain at the final max pooling layer. The pooled feature map is finally flattened and fed into the fully connected layers. In the first part of this study, the effectiveness of a pre-trained VGG-16 network that was previously trained over the 'imagenet' dataset with thousands of classes of images is accessed for forgery detection in signatures. The fully connected layer i.e. the top layer of pre-trained VGG-16 model normally generates 1000 different output labels, but the chosen problem in this study is a binary classification work. So instead of entirely utilizing the architecture of pre-trained VGG-16 network, we have customized the network with a new top portion i.e. the fully connected layer and output layer by freezing the pre-trained convolution layers and unfreezing the top fully connected layer of original VGG-16 network. The pre-trained convolution layers of VGG-16 will help to convolve visual features from the input preprocessed signature images. Then the non-frozen pre-trained layers are trained with the collected signature images. In the next part of the study, instead of utilizing the pre-trained weights of the convolution layers, the entire VGG-16 model with new top portion is trained with the signature dataset and its classification performance is accessed.

## III. EXPERIMAENTAL RESULT DISCUSSION

A computer system with an AMD Ryzen 5 3600 CPU clocked at 4.2Ghz, Nvidia RTX 2060 6GB GPU(Graphics Processing Unit) and 16GB of 3200mhz RAM was used to implement the suggested ASVS. The study made use of Python 3.9.5 with the OpenCV image processing library, Tensorflow version 2.10, Matplotlib and SK.learn library.

The handwritten signatures were retrieved from the freely accessible CEDAR dataset. There are 2640 grey scale signatures altogether in this dataset, 1320 of which are real and 1320 of which are forgeries. We have also added 60 images of each real and fake signature images of our own, making the total signatures for the dataset to be 2760. Then all the images are rescaled to a size of 250X250. The original and preprocessed image is shown in **Fig. 2**. Then the dataset is divided in a proportion of 80:20 and the preprocessed images from the training set are given as input to the VGG-16 model. Initially, adapting the transfer learning mechanism, the pre-trained weights of the VGG-16 are used in the model allowing

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only the end fully connected layers to be trained. Next the performance of the trainable VGG-16 model, which is entirely trained with the training set employing the SGD optimizer keeping the learning rate to 0.001 and considering binary_crossentropy as the loss function, is observed. During training, the epoch size was set to 20, batch size to 30. The accuracy and loss graphs of related to the two DNNs are shown in **Fig. 3** and **Fig. 4** respectively. The training and validation accuracy obtained by the two VGG-16 based ASVS are presented in **Table 1. Fig. 5** and **Fig. 6** shows the confusion matrix obtained by both ASVS for the test set.

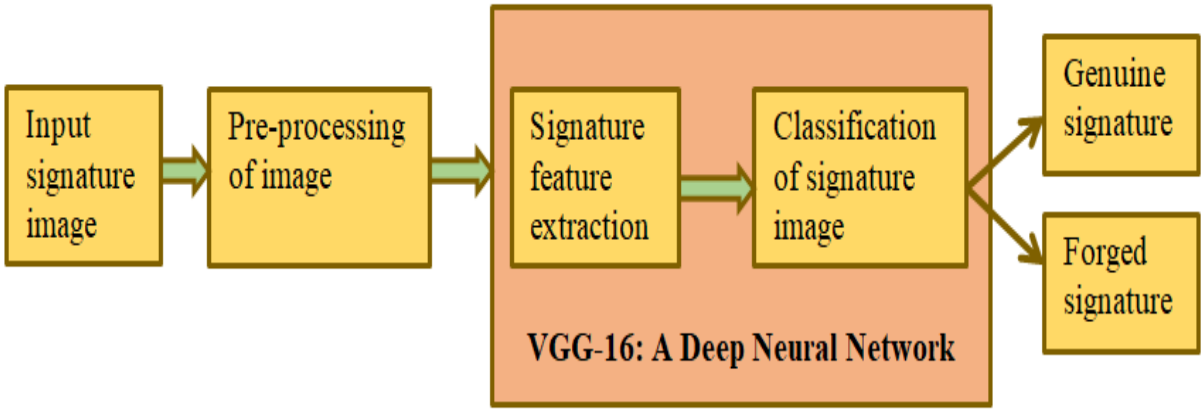



**Fig.1**. Working Flow Diagram of the VGG-16 based ASVS.

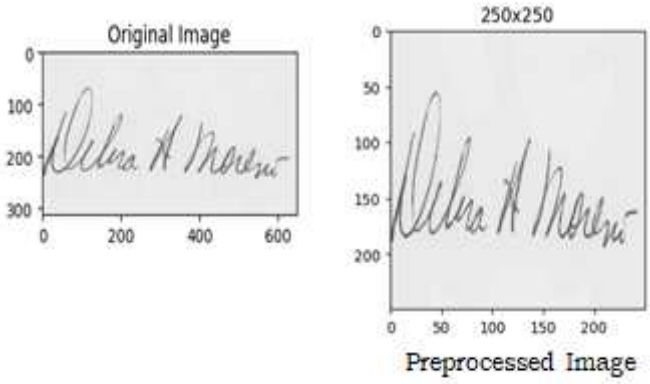



**Fig. 2** Original with the pre-processed signature image

**Table 1.** Training and validation performance comparison of the two VGG-16 based ASVS

| Model | Training accuracy | Validation accuracy |
|---|---|---|
| Trainable VGG-16 | 0.997 | 0.959 |
| Pre-trained VGG-16 | 0.802 | 0.769 |

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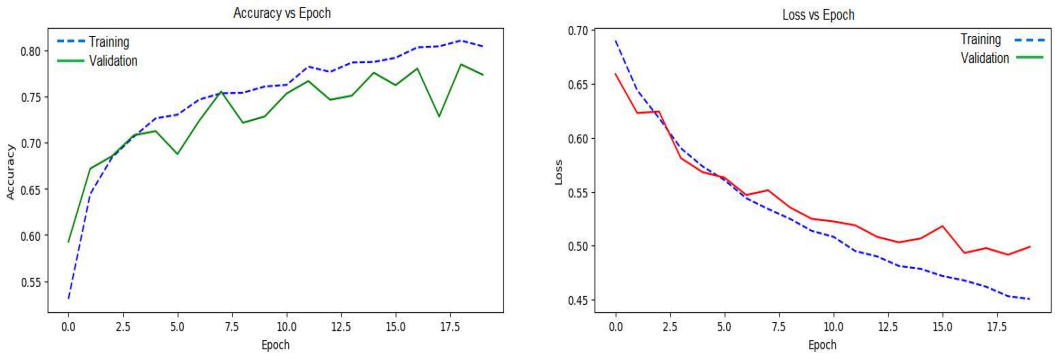



**Fig. 3** Accuracy and Loss obtained during training and validation phase of pre-trained VGG-16 based ASVS model

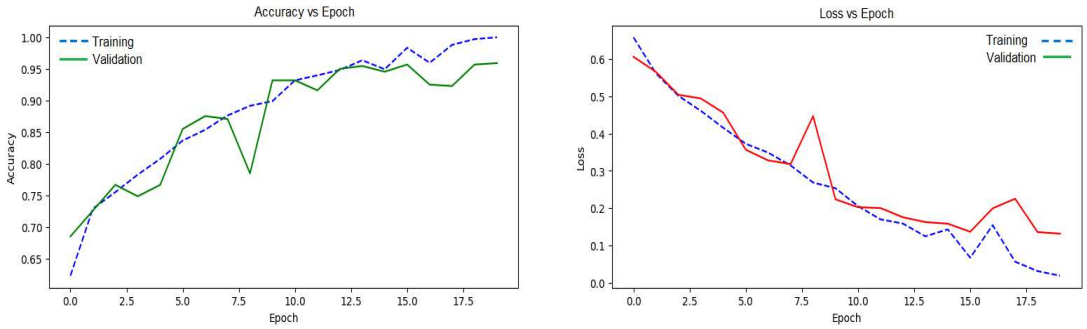



**Fig. 4** Accuracy and Loss obtained during training and validation phase of trainable VGG-16 based ASVS model

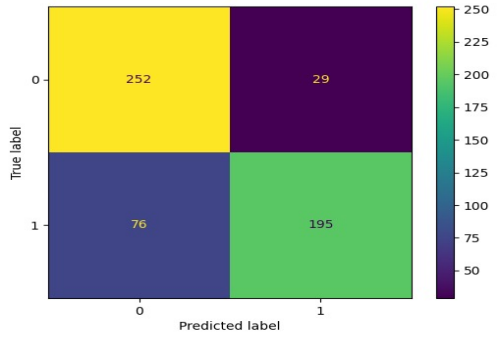



**Fig. 5** Confusion matrix related to test set (Pre-trained VGG-16 based ASVS)

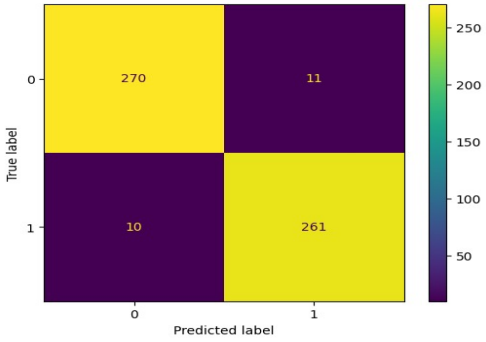



**Fig. 6** Confusion matrix related to test set (Trainable VGG-16 based ASVS)

Analyzing the outcomes depicted in Table 1 and Figs. 3 to 6, it is clear that the trainable VGG-16 model provides better accuracy during training, validation and test phase compared to the pre-trained model. The comparison of the two models related to test data is also presented in **Fig. 7**.

Finally the prediction outcome of the model for two random images one belong to each class is depicted in **Fig. 8** that was obtained by using the Model.predict () function from the tensorflow library. the Model.preditct() returns a tuple of probabilistic predicted value in terms of (X,Y), where X indicates the probability of the image belonging to class 0, which is forgery in this case, and Y indicates the probability of the image belonging to class 1 which is labeled as genuine for this case. Comparison between the two probabilistic values was made. If X was found to be greater than Y, it implied that the image belonged to class 0, the forgery class, and if Y was found to be greater than X, it implied that the image belong to class 1, the genuine class. The Model prediction returns the probabilistic tuple [X,Y] with corresponding values [1.0679432e-06, 9.9999893e-01] for the case of a random image as shown in **Fig.8 (a).** Value of Y was found to be greater than X which meant the signature image is a genuine signature. Additionally, the random image shown in **Fig. 8(a)** is searched in the CEDAR dataset manually is found in the 'full_org' directory named as 'genuine_29_12' which further confirmed that the prediction made by the model is indeed correct. Similarly the model prediction returns the probabilistic tuple [X,Y] with corresponding values

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[[9.9977905e-01, 2.2100362e-04]] for the case of a random image shown in **Fig. 8(b).** Value of X was found to be greater than Y which meant the signature image was a forged signature. When, the random image shown in **Fig. 8(b)** is searched in the original dataset manually, it is found in the 'full_forg' directory named as 'forgeries_19_17' which further confirmed that the prediction made by the model is indeed correct.

## IV. CONCLUSION

In this research, verification of signatures and identifying the forged one among them is addressed as a binary classification task. The key point is the comparison of the attainment of the pre-trained and trainable version of VGG-16 model for the same task. By conducting experiments on the well-known CEDAR dataset and examining the outcomes, it is clear that the trainable VGg-16 based ASVS is better able to identify forged images from the image pool than the pre-trained model. To further advance this research, data augmentation can be adopted to yield a massive synthetic dataset for improving the performance of the model. Another research direction that we plan to perform is to design a less complex CNN model with fewer parameters to achieve higher accuracy in detecting forgery in both online and offline handwritten signature verification process.

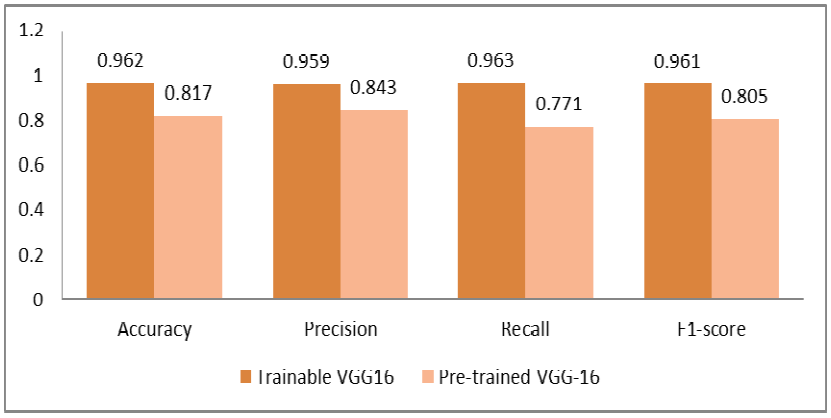
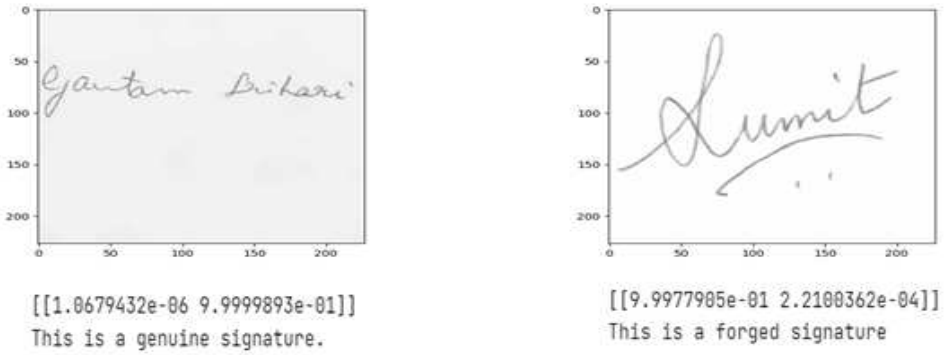



**Fig. 7** Comparison of test outcome related to both VGG-16 based ASVS




**(a)** **(b)**

**Fig. 8** Classification of a random image from the test dataset as genuine or forged

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